

Ten questions concerning renewable energy communities

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ABSTRACT

Renewable energy communities (RECs) represent a transformative approach to achieving sustainable energy systems at local levels. By integrating various renewable energy sources and advanced technologies, RECs can efficiently target energy self-sufficiency and endorse renewable distribution while promoting environmental, economic and social benefits. Despite their increasing relevance, the complexity of developing and operating RECs, as well as many of the management aspects regarding the distribution of shared energy and, therefore, the economic gain, necessitate comprehensive modelling and strategic planning. This paper addresses the multidimensional aspects of RECs through a structured approach based on ten pivotal questions, covering REC definition, technological components, optimization and simulation methods, monitoring and management strategies, business model and policies, and future urban trends such as Positive Energy Districts. The questions were selected considering a logical structure of all fundamental aspects and progressing from the initial state to the current and future of the RECs, based on an accurate assessment of the regulatory contexts, the supporting policies, the scientific literature and the results of case studies and carried out projects. The analysis highlights that RECs contribute significantly to achieving climate and social goals, yet their diffusion is deeply influenced by the regulatory framework, but economic incentives and citizen involvement also play a significant role for their success.

1. Introduction

Given the urgent challenges climate change poses and its related environmental impacts, global initiatives are focused on implementing significant changes within the energy sector. Significant efforts are aimed at reducing greenhouse gas (GHG) emissions, increasing the share of energy from renewable energy sources (RESs), and enhancing energy efficiency [1]. In recent years, transitions to clean energy have accelerated strongly, driven by government policies and industrial strategies which have promoted actions for greater energy independence, as a response to geopolitical instability caused by open conflicts around the world and soaring energy prices [2]. In this framework, the Agenda 2030 for Sustainable Development marked key goals essential for the future development of humanity and the planet [3]. Concurrently, the European Green Deal lays a solid foundation for achieving net zero emissions by 2050 and reducing GHG emissions by at least 55 % by 2030, compared to the pre-industrial levels [4]. To achieve climate neutrality, it is essential to explore various technological pathways, including the electrification of consumption, the integration of hydrogen

technologies, and the implementation of CO₂ capture techniques [5]. The energy transition also requires citizens to take a central role by developing greater awareness of their energy choices and actively acting towards achieving global targets [6]. Significant changes in the energy consumption and production paradigm, where RESs are increasingly replacing fossil fuels, are flanked by energy sharing, which is emerging as a key factor for decarbonisation [7]. In this context, Renewable Energy Communities (RECs) represent an energy model that encompasses all the key aspects of the energy transition. These projects bring together a diverse range of stakeholders who collaborate in sharing energy generated from a RES-based plant. This partnership is designed to promote environmental sustainability while also fostering economic growth and enhancing social well-being. The local aspect of these configurations makes it possible to guarantee effective spillovers on the territory and to encourage the active participation of members by embracing the community identity [8]. The worldwide spread of energy-sharing configurations varies based on their adaptation to existing regulatory frameworks. From this perspective, the formal introduction of RECs into European legislation through the Renewable Energy Directive 2018/2001 (RED II) [9] has been a cornerstone for their

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Nomenclature			
AC	Allocation coefficient	MV	Medium voltage
ACO	Colony optimization	NLP	Non-linear programming
ADR	Automated demand response	P2G	Power-to-gas
BESS	Battery energy storage systems	P2H	Power-to-heat
CEC	Citizens energy community	P2P	Power-to-power
CHP	Cogenerator	P2V	Power-to-vehicle
CSC	Collective self-consumption	P2X	Power-to-X
DHN	District hot network	PED	Positive energy district
DSF	Demand-side flexibility	PG	Power Grid
EEG	Erneuerbare-Energien-Gesetz	PSO	Swarm optimization
ERDF	European Regional Development Fund	PtP	Peer-to-peer
EU	European Union	PV	Photovoltaic
EV	Electric vehicle	PVT	Photovoltaic thermal
GA	Genetic algorithms	QP	Quadratic programming
GHG	Greenhouse gas	REC	Renewable energy community
HP	Heat Pump	RED II	Renewable Energy Directive
HV	High voltage	RES	Renewable energy source
JRC	Joint Research Centre	RESS	Renewable Electricity Support Scheme
LP	Linear programming	SDG	Sustainable Development Goal
LV	Low voltage	SME	Small and Medium Enterprises
MILP	Mixed-integer linear programming	V2G	Vehicle-to-Grid
		VPP	Virtual Power Plants

implementation. The diffusion of RECs has made it possible to evaluate their potentialities and the main aspects concerning organizational, technical, and multidimensional components in more and more detail. The comprehensive understanding of many aspects of these configurations provides a thorough overview. Many aspects have indeed been regulated by the national transpositions of the member states, although there are similar or different implementations for each of them depending on their compliance with the existing regulatory framework [10]. In this regard, rules have been imposed regarding the type of members to be included, the maximum size of the configuration and the energy sharing scheme. Different economic compensation mechanisms have been developed depending on the energy shared, generated by RES plants, or specific objectives achieved [11]. The application of business models for RECs allows optimal management of economic benefits together with ideal governance of membership aggregation for an operational sustainability and long-term scalability [12]. The spread of RES installations has been a driving force for the spread of RECs, which in turn have supported the installation of new plants as enablers of the energy transition. Solar photovoltaic (PV) and wind turbines are today the decentralized technologies with the largest installed capacity [13], but since the associated source is not programmable, they are often flanked by other more programmable technologies. In the context of RECs, the implementation of storage systems, the use of other energy carriers such as hydrogen or the application of specific strategies allows maximising energy availability. Their implementation maximises the REC flexibility in order to reduce dependence on the power grid (PG) and increase energy self-consumption [14]. However, the complexity of energy flows requires the application of dedicated optimization and simulation techniques in order to manage REC performance optimally. Efficient data monitoring strategies are essential for the proper functioning of the configuration and for achieving the associated environmental, energy and social benefits.

The aim of this paper is to present a comprehensive overview of the essential aspects of RECs. In particular, technological components, optimization and simulation methods, management and monitoring strategies, as well as policy, business models and future directions have been explored, pointing to provide a toolkit for interested stakeholders.

2. Methodology

RECs play a crucial role in the energy transition process, but addressing the various facets of this topic effectively requires a comprehensive analytical methodology capable of highlighting each aspect to maximize the benefits of their implementation. The methodology employed to structure this paper is based on a thorough and critical analysis of the key factors associated with the concept of RECs. The selection of the ten questions at the core of this paper results from an extensive literature review, followed by a detailed evaluation of the main research findings in recent years. Each question has been carefully chosen not only to provide a general overview of the topic but also to delve into key aspects that are essential for understanding and guiding REC implementation, while also pointing toward future directions. The structure of the paper follows a logical and progressive flow, starting with foundational definitions and advancing to more complex topics such as sector coupling, flexibility and upscaling strategies, including economic, environmental and social dimensions. A focus was placed on the issues surrounding the definition of RECs, including the types of members that can form these communities and the resulting configurations that can be achieved. A crucial factor in this analysis is the level of compliance with each country's regulatory framework, which significantly influences REC implementation. To ensure precise and well-founded responses, a thorough evaluation of the existing legislative frameworks was conducted, accompanied by the analysis of relevant case studies from the literature. The technological aspect of RECs is another central factor, based on the latter: RES-based installations and commonly used strategies provide valuable insights into their operation and optimization. To effectively achieve the objectives set at the national, European, and global levels, it is essential for RECs to adopt technical optimization and energy flow management approaches. These objectives can be effectively pursued through the integration of simulation methods and flexibility solutions, supported by tailored tools and strategies. Optimal management of energy-sharing configurations allows for tangible benefits, from the local to the global level, enabling projects to scale to broader areas, supported by the policies in place. The analysis of the scientific literature on RECs was instrumental in shaping the responses to the ten proposed questions. However, only a coherent vision of the topic allowed to effectively integrate the legislative context

with the outcomes derived from the case studies examined. This methodological approach enables to provide comprehensive and well-rounded responses to each issue, offering the flexibility to explore each aspect in detail or to cluster related topics as needed, depending on the relevance and circumstances of each particular dimension.

3. Ten questions

3.1. What and who constitutes a renewable energy community?

Answer: RECs, legally defined by the RED II, are voluntary association of individuals, SMEs and local authorities, aiming to promote a fair and sustainable energy transition by integrating renewable energy sources and fostering social, economic and environmental benefits. Their success, however, depends on overcoming challenges such as the selection of members with complementary energy profiles, high investment costs, technical complexity and regulatory barriers.

RECs were first legally defined in the European regulatory framework in 2018 through the RED II [9]. They are legal entities based on the voluntary participation of stakeholders, who can be natural persons, small and medium enterprises (SMEs) or local authorities. However, various other member types are involved in implementing RECs, which may differ based on the pursued aim and the specific circumstances. This heterogeneity plays a pivotal role in achieving the environmental, energy and social objectives of these projects. The participation in RECs can be influenced by several key factors, including social, economic, technological and policy aspects. RECs serve as an instrument for implementing a fair energy transition that involves actors at a local level and makes them actively involved in decision-making processes, regardless of their individual financial capacity [15]. Individuals are driven by values of sustainability and technical skills, which unite with local identity and social cohesion felt at a community level [16]. Additionally, the potential for economic savings on energy costs remains a significant attraction both for experienced stakeholders and consumers [17]. Despite the strong social drivers supporting the development of RECs, challenges such as high investment costs and operational complexities continue to hinder progress. Addressing these issues requires robust policy support and innovative financial solutions [18].

The social dimension of the energy transition process is mainly emphasized with the participation of residential users as members of the REC [19], providing concrete opportunities to reduce energy poverty conditions for included low-income households [20]. The participation of residential users allows for load shifting through the implementation of home energy management systems, which helps optimize the use of energy produced by RES-based plants [21]. The community approach presents benefits for residential buildings equipped with RES plants, retrofitted through the replacement of existing energy conversion systems [22]. Indeed, introducing heat pumps (HP) as energy conversion systems to meet space heating demands led to enhancing the thermal sector's flexibility. This approach taps into non-electric loads, enabling maximum energy sharing through the power-to-heat (P2H) approach [23]. Simultaneous control of indoor comfort conditions for residential users of a REC through the air-conditioning systems and batteries associated with the PV system results in economic savings of up to 59 % compared to the case where users act individually [24]. Additional cooperation among community members in demand response events, besides supporting energy balance, results, as a consequence, in economic savings both for individuals and the community [25]. However, due to the current lack of infrastructure for measuring energy loads, non-intrusive techniques are needed to assess them and maximize self-consumption within the community [26].

Electric vehicles (EVs) integrated into a REC represent additional electrical loads that can be optimised. This integration represents an opportunity for the transport sector of real decarbonisation since the electricity used comes mainly from RES. EVs' charging flexibility blends with the load shifting of appliances [27], and in specific applications, the

RES plant's surplus can be offered to membership to external EV owners [28].

The advantages of a REC can be enhanced by exploring various user combinations [29]. One important technical factor in selecting members optimally is their energy demand [30]. In fact, the diversity among different energy demand profiles—such as residential, industrial, tertiary, and agricultural—can help align the energy demand with the generation patterns of the entire community [31]. For instance, a multi-purpose REC that includes six residential buildings and two restaurants, that show different energy load profiles, demonstrates significant primary energy savings, reduced environmental impact, and economic benefits when compared to base configurations where the same users draw electric energy from the PG [32]. Moreover, incorporating an industrial user positively influences the energy-sharing indicators of a REC that consists of municipal and residential users [33]. The participation of SMEs, indeed, presents great potential for a REC, due to their energy demand and propensity to contribute to decarbonisation [34]. highlights that SMEs favour the involvement of local suppliers, in line with the RED II aim of providing benefits to the territory where the REC operates [9]. A moderate administrative effort and a strong interest in demand-side management and peer-to-peer trading have a positive influence, unlike an excessively lengthy return on investment time that negatively impacts SMEs' involvement. On the other hand, matching through municipalities boosts private citizens' investment in RECs [35]. Public entities included in the REC have a positive effect due to the availability of areas for the installation of RES-based plants on the buildings and the organizational and financial resources, as well as consumption data used for model's validation [36]. Both public and private energy players recognise among the main barriers to participate in a REC, planning and legal framework followed by the lack of resources and expertise [37]. The interest in clean energy and new energy production and consumption paradigms has led to important transformations in the communities' sphere that have increasingly developed an interest in energy aspects, focusing on RES projects and energy governance, extending to social, political and economic objectives. In this context, types of energy communities such as municipal, rural or island ones have been implemented [38]. In municipality-driven REC, the local municipality acts as a promoter of the project involving other actors with the aim of offering positive consequences at the local level [39]. Rural energy communities gain more attention due to important technological, economic, and social factors for their growth and sustainability. Studies show that key priorities for rural energy advancement encompass community involvement and the integration of agriculture. Current trends reveal a shift from localized solutions to integrated hybrid systems and smart grids [40]. Small islands that are not connected to the PG represent ideal locations for developing projects aimed at energy independence and self-sufficiency, including RES-based plants and storage solutions [41]. Optimal energy management on these islands requires a careful design of plants aimed at maximizing energy production and efficient use, minimizing costs related to unnecessary systems [42]. The feasibility of islands' RECs can be achieved by integrating significant energy loads that remain consistent regardless of the seasonal fluctuations in tourism. For example, incorporating a water desalination unit can enhance the economic viability of the REC by allowing for the utilization of surplus energy from PV plants [43]. The RECs can also facilitate the decarbonisation of sectors still heavily dependent on fossil fuels. In this regard, ports are complex and very energy-intensive realities, to which are linked numerous electricity demands for services at the dock, mooring or energy distribution for power boats, which can be partially met by the installation of community RES plants and battery energy storage systems (BESS) [44]. The port authority is seen as the primary stakeholder that, as a public body, can promote and facilitate the establishment of one or more RECs, involving maritime stakeholders [45]. Table 1 summarises the main factors which favour or restrict participation in a REC for each of the considered members or organisations types.

Table 1

Positive aspects and barriers for the participation in a REC of each member typology.

REC member affinity	Facilitations	Barriers
Residential	Propensity for social aspects, flexibility loads	Lack of load profile measurement
SMEs	High and constant energy demand, propensity to decarbonization	High return time on investment
Public entities -Municipality-	Area availability for plant installation, organizational and financial resources, energy consumption data	Legal framework, lack of expertise
Rural	Technological, economic, and social factors for their growth, propensity to sustainability	Community involvement and integration of agriculture
Island	Energy independence	Not PG connection, seasonal energy loads
Ports	High energy demand	Complex organization

3.2. What is the current status of RECs' development worldwide?

Answer: RECs, or generally energy sharing models, are increasingly recognized worldwide with their development shaped by diverse regulatory frameworks, from Japan to South Korea, the U.S. and the European Union, where the Clean Energy for All Europeans Package and national transposition of RED II introduced physical and virtual models for energy sharing, as well as incentives and proximity rules.

The RECs embody important realities that are recognized worldwide, contributing significantly to the current energy system. Their development has adhered to various operational modalities and regulations, reflecting the unique regulatory environments of each country. The necessity of energy sharing models that cater to local requirements has been greatly enhanced by the increasing installation of RES-based plants, particularly PV systems, as well as the simultaneous rise of prosumers in the energy landscape. In Japan, these two incentive factors have been supported by a reduction in installation costs and the expansion of RES in previously less favourable contexts. However, this progress has required a solid regulatory framework of support [46]. Instead in Russia, microgrids for power supply significantly alleviate rising electricity costs for industrial and commercial consumers, ensuring reliable, environmentally friendly power supply. Here, the implementation of the energy community is based on these microgrids, and a regulatory framework was developed to govern their operation, focusing on business practices and enhancing efficiency in remote areas [47]. South Korea aims for sustainability by promoting net-zero-energy buildings and decarbonizing existing communities. The maximization of energy from RES plants is achieved by retrofitting buildings and energy sharing, achieving a positive energy balance [48]. In order to accelerate the energy transition at the national level in the United States, regional and localized entities started the development of focused projects to achieve the target of 100 % renewable energy [49]. In Utah, the Community Renewable Energy Act has been enacted, leading to the establishment of the Community Renewable Energy Agency. This initiative will allow up to 23 communities in the state to achieve the goal by 2030 [50]. RECs play a crucial role in each country's energy transition by providing cooperative benefits from renewable generation facilities and potential opportunities from consumption aggregation for energy efficiency services [51]. To effectively implement these advantages, a strong legislative framework is necessary to establish clear applications and regulations governing their operation.

The European Union (EU) introduced the Clean Energy for All Europeans Package as a comprehensive legislative framework designed to support the energy transition. Its main goals include reducing GHG emissions, increasing the use of RESs, and enhancing energy efficiency. In this context, the package emphasizes the importance of placing

citizens at the centre of this transition by introducing concepts such as active consumers, individual and collective self-consumers (CSC), citizens energy communities (CECs) and RECs [11]. European directives introduced general concepts for each of these entities, but each member state has, through transposition into national law, defined the application and the implementation rules in accordance with its own legislative context [52].

Regarding RECs, the primary distinction is found in how the advantages of collective self-consumption are reflected in the members' energy bills. Specifically, two models have been established in the European setting: virtual and physical. The latter entails a direct energy bill discount for community-shared energy by each member, who pays for the remaining withdrawn energy, while a different internal price valorizes the energy traded within the community [10]. Spain and Portugal consider this approach for RECs and employ allocation coefficients (ACs) to assign shared energy to each member. Portuguese regulation allows the use of dynamic ACs that can be assigned by REC members or by a designed Management Entity [53], while Spanish regulations involve static ACs, fixed over a specific time period [54]. Moreover, in Spain, RECs members have to respect the maximum distance of 2 km from the generation unit [55]. The Belgian regulation also considers a physical scheme, and the economic incentives are guaranteed as a discount on members' bills based on shared energy on a sub-hour basis [56]. In Belgium, the regulation is not uniform throughout the country and has been developed separately for the Brussels capital region, Flanders and the region of Wallonia [52]. On the other hand, the virtual scheme establishes no energy bill discount for REC members, but shared energy results in economic incentives according to the country's rules. In accordance with the Italian regulatory framework, economic incentives are granted to REC depending on the price of the electricity market zone where the configuration is located and the size of the RES-based plant, in addition the members should be connected to the same high/ medium voltage (HV/MV) electrical substation [57]. Economic revenues can be divided among members or, according to the REC's statutes, used for sustainability or social purposes. The Netherlands is also considering a virtual scheme for energy communities. The subsidy depends on a fixed and variable component, accounting for fluctuations in energy prices. However, the incentive is calculated based on the energy produced by community plants, and the shared energy is not measured [58]. Germany also presents economic offsets for RECs, through various programs, including incentives, compensation or low-interest loans. Although the Country has been a pioneer in developing collective configurations to exploit the energy produced by a RES plant, the regulatory framework does not fully comply with the requirements established by the RED II [59]. In Germany, indeed, energy communities consist of renewable energy assets owned by individuals, small businesses, or cooperatives. These entities can also sell surplus energy to nearby end-users [39]. The Austrian regulatory contest provides that RECs may produce, consume, store or sell energy from RESs. To encourage participation, a reduced distribution grid tariff is applied to electricity traded within the REC. Additionally, participants benefit from the elimination of all electricity taxes and green energy charges [60]. The concept of proximity for RECs outlined in the RED II, in Austria results in a connection between members and generating units via low or medium-voltage (LV/MV) grids, respectively for local or regional level configurations [61]. In France, the REC provides for energy sharing between producers and consumers following CSC. The maximum distance among members can be 2 km if they are connected to the LV grid, but it can reach up to 20 km with specific approval for low density population areas [52]. Not all European member states transposed RED II into national laws, however, projects using renewable sources and energy use are encouraged at community level. In Ireland, the Renewable Electricity Support Scheme (RESS) supports energy production from RES and provides priority mechanisms for community-based projects [52], including heat sharing. RECs can access the electricity market and receive economic benefits by

competing with other players [37]. Key features of the RECs for the investigated European member states are shown in Fig. 1.

3.3. What are the key renewable technologies in RECs?

Answer: Solar PV technology, often complemented by storage systems, is the most commonly adopted RES in RECs, due to their affordability, modularity and ease of installation. Hybrid solutions with wind, biomass, and hydropower are increasing with the additional benefit of advancing Power-to-X strategies.

Solar PV plants represent the most widely used RES technology, due to its modularity, ease of installation, and relatively low investment costs compared to other systems [62]. Over the years, there has been a significant reduction in the price of solar PV systems, which has facilitated the installation of plants for citizens. In this context, also energy sharing configurations have primarily utilized PV technology in community projects, leading to the development of so-called solar energy communities [63]. These installations are typically set up on the rooftops of buildings [64] or in dedicated areas, depending on specific project requirements. A multi-tower concentrating solar power plant requires, for example, specific area availability while offering substantial benefits in terms of high yields of electricity generation [65]. The exploitation of solar energy can also be used to meet thermal energy needs by installing thermal solar collectors or photovoltaic thermal (PVT) systems, which can simultaneously satisfy both electrical and thermal energy demands [66]. Despite the benefits of solar technologies, they may not be the most suitable option for all regions, such as those with greater availability of other RESs, as Denmark, where RECs are mainly using wind [67].

However, these sources have several limitations, mainly related to the non-programmability of production which depends on source availability that can vary seasonally or follow specific daily production patterns.

More flexibility can be offered by integrating storage systems as BESSs to overcome these barriers, despite their significant economic

impact [68]. The optimal management of BESS coupled with PV considering energy prices and making forecasts of weather data from which to assess the producibility of the plant, however, allows for obtaining substantial economic revenues and the minimization of associated operating costs [69]. Their use involves a control strategy that has to fit the community's demands [70], which, if optimized, leads to greater independence from the PG and more incentives on energy sharing [71].

On the other hand, hybrid solutions that combine more RES systems, such as PV and wind turbines, can offer additional advantages [53], also maximized with the integration of other greater programmability technologies. These include biomass-fuelled plants that exploit a locally available resource with consequential significant employment impacts on the territory [72]. The combination of a biomass-fuelled cogenerator (CHP) and existing PV system simultaneously supplies the thermal and electric energy demand of REC members [73]. Specific crops can be exploited for the production of alternative biofuel pellets to cover the thermal demand of a REC [74]. The work in [75] presents a REC where woody products from local activities, such as sawmills, are used to activate a CHP, which covers community demand coupled with a small hydropower plant and PV system. Depending on the specific availability, geothermal or marine renewable energy technologies can also be used, even in conjunction with less complex systems such as PV and wind systems [76]. Adopting hybrid configurations that integrate several RES technologies and storage systems presents a promising opportunity to enhance the self-sufficiency of the CER while effectively reducing PG withdrawals [77].

Greater flexibility can be achieved in REC that, in addition to BESS, also predisposes thermal energy storages [78], allowing thermal energy sharing among members through a district hot network (DHN) [79]. Additionally, flexible loads such as EVs and heat pumps [80] can be inserted, or RES energy surplus can be stored in green hydrogen, also used for Fuel Cell EVs [81].

Different Power-to-X (P2X) strategies can be applied within a REC in order to improve self-consumption of energy, including P2H, power-to-

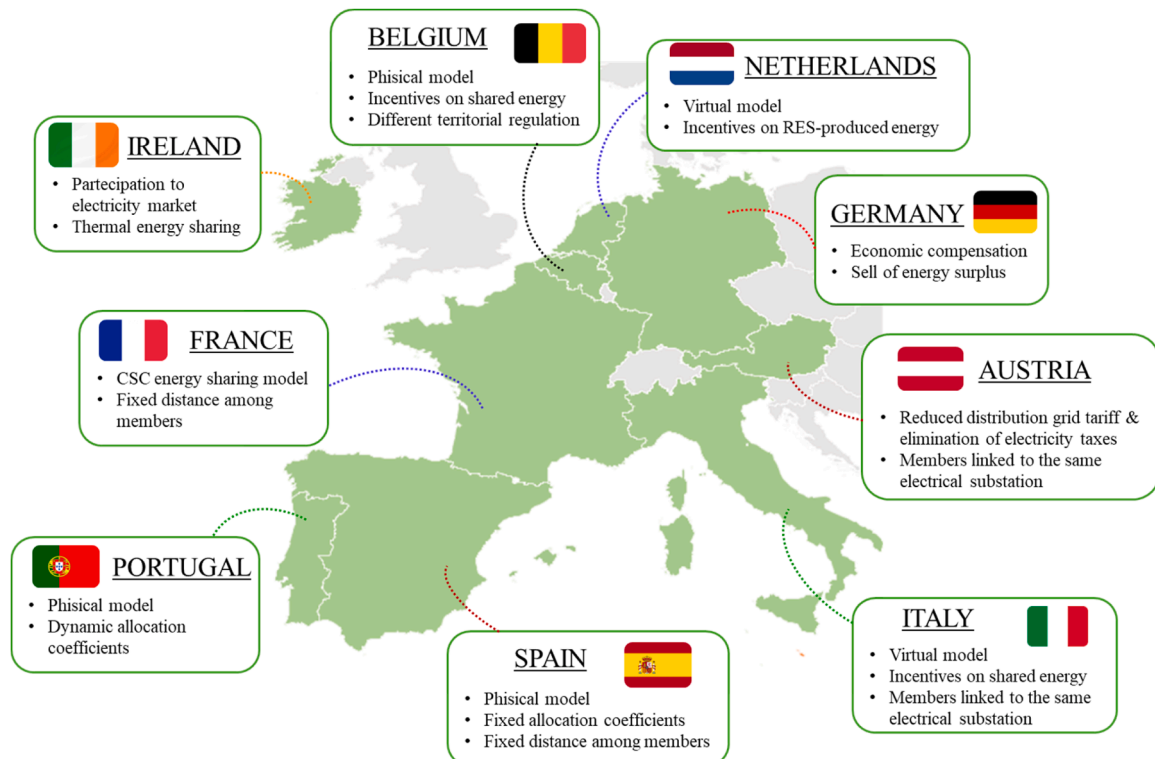


Fig. 1. Key features of some EU States RECs regulation.

vehicle (P2V), power-to-power (P2P) and power-to-gas (P2G) [82].

Finally, hydrogen is an emerging possibility in recent years whereby RES surplus energy can be used via electrolyzers for the electricity postponed production [83], promoting Renewable Hydrogen Energy Communities [84]. These systems are also part of the PtP approach. Fig. 2 presents the most diffused RESs, in relation with energy carriers and consequences strategies application in the RECs field.

3.4. How can optimization and simulation approaches act RECs development?

Answer: The operation and development of RECs require tailored optimization and simulation frameworks to manage the complexity of energy flows, storage, and market interactions. Usually, optimization models are used to design and operate RECs, while simulation approaches are adopted to assess system behaviour under varying conditions.

As emerge from the definition of the RED II, the operation and development of RECs require tailored modelling frameworks to take the complexity of energy flows, variable renewable generation, storage integration, and market interactions into account. To this end, optimization and simulation approaches play a crucial role for the design, implementation and performance evaluation of RECs.

Optimization techniques are implemented to determine the most efficient way to operate, design, and expand RECs by minimizing costs, emissions, or energy imports while maximizing self-consumption, energy efficiency, or economic returns. On the other hand, simulation approaches enable the assessment of REC performance under different operational scenarios, boundary conditions, and policy frameworks. Optimization and simulation tools have therefore important differences and are employed differently depending on the main purpose of the study.

Starting with optimization models, they are commonly implemented to design and manage RECs efficiently. These models are applied for a range of objectives, including energy flow management, e.g., minimizing grid dependency while maximizing self-consumption and local distribution among REC members [20] sizing of renewable generation and storage systems [85], economic optimization, e.g., including tariffs scheme and revenue models [29], energy sharing mechanisms [56], investment and operation costs [86], emission reduction [87], demand-response and other flexibility solutions [88].

In a REC, the optimization problem typically aims to minimize costs, emissions, or energy imports while maximizing self-consumption, flexibility, and sustainability. A general formulation for optimization problems in RECs can be expressed as:

$$\min F(x) = [F_1(x), F_2(x), \dots, F_n(x)] \text{ (objective function)}$$

$$g_i(x) \leq 0, \forall i \in \mathcal{I} \text{ (inequality constraints)}$$

$$h_j(x) = 0, \forall j \in \mathcal{J} \text{ (equality constraints)}$$

$$x \in \mathcal{X} \text{ (decision set)} \quad (1)$$

where $F(x)$ is a vector of functions representing conflicting objectives (e.g., cost minimization, emissions minimization, self-consumption maximization), each expressed by $F_i(x)$, $i = 1, \dots, n$. The decision variable x may include energy-related and storage operation parameters or investment levels, for example. The set of inequality constraints $g_i(x)$ captures technical and regulatory limitations such as generator capacities, tariffs or emission. The equality $h_j(x)$ typically model energy balance equations, e.g. supply should equal demand at each time step, storage continuity equations, or thermal-electrical coupling. Finally, the feasible region $\mathcal{X}$ encompasses discrete, continuous, and binary decision sets, reflecting technological or policy related restrictions.

Several optimization techniques are applied in RECs' modelling. The most diffused belong to mathematical programming, such as linear programming (LP) [89], mixed-integer linear programming (MILP) [90] quadratic programming (QP) [91], and non-linear programming (NLP) [92] commonly used for determining the optimal energy dispatch as well as the financial benefits deriving from energy sharing within the REC, and investments decisions. Meta-heuristic algorithms and approaches [93], such as genetic algorithms (GA) [94], swarm optimization (PSO) [95], ant colony optimization (ACO) [96], or commercial software, such as HOMER [97] or TRNSYS [98] are able to handle complex, non-convex problems. When there is the need to incorporate uncertainty in modelling, such as the one coming from RES generation, demand fluctuations, and market conditions, stochastic optimization techniques are implemented, given their capability to deal with chance-constrained programming [99]. RECs may combine different sectors and aim to achieve opposite objective. To being able to balance trade-offs between costs, emissions, and energy efficiency, Pareto-based

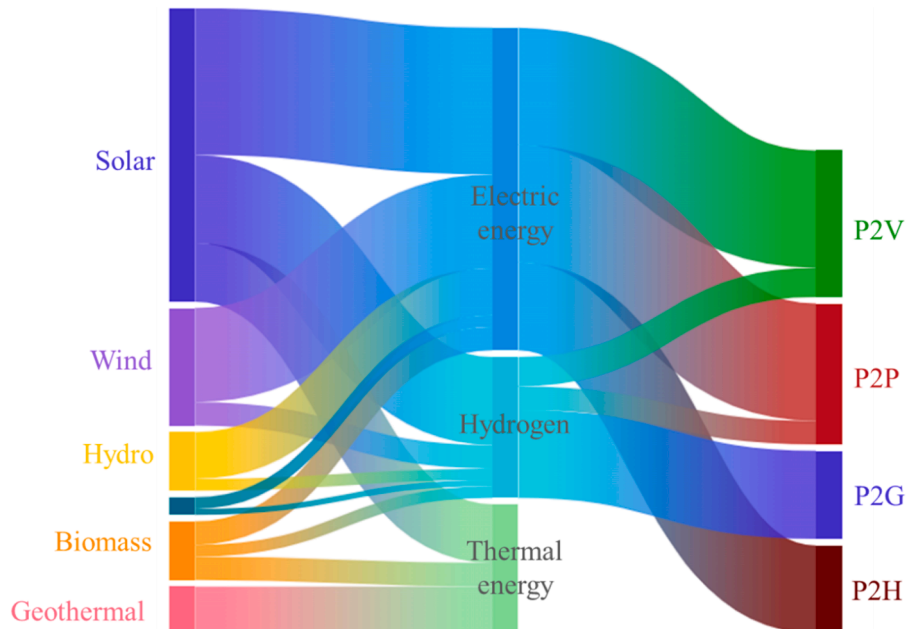


Fig. 2. Most diffused RESs, energy carriers and application strategies in the RECs.

approaches are preferred, with the formulation of a multi-objective function [66]. Recently, AI techniques and deep learning [100] are going to be exploited and implemented, to dynamically optimize energy flows in real-time [101] and to guarantee control mechanisms [102]. These methods can be then combined to obtain hybrid approaches [103], training to couple the benefits and advantages of different methods together. A comparison of the so-far discussed optimization methods, with a focus on pros and cons and most related application in the context of RECs is provided in Table 2.

Regarding simulation approaches for RECs, these are employed to assess system behavior under different scenarios and can also be used to complement optimization methods. In particular, simulations enable performance evaluation in terms of energy, economic and social impacts before implementation [104] sensitivity analysis to understand the impact of variations in demand, generation, and tariffs [105] or impact assessment [106] or future predictions. Among the most implemented simulation methods, agent-based models (ABM) consider REC members as autonomous decision-making entities [107].

Unlike optimization, simulation approaches typically do not follow a unified mathematical formulation as they are based on the modelling of the behavior of systems over time, given a set of input data, decision rules, and system interactions. However, there are some widely adopted mathematical structures, especially for the most common modelling techniques, such as ABM, in which the simulation consists of a set of autonomous agents, e.g. prosumers, grid operators, aggregators, that follow specific decision rules and interact within the REC. Each agent can be characterized by a state $S_i(t)$ and behavior rules B_i updated at discrete timesteps. A generic ABM formulation can therefore be:

$$S_i(t+1) = B_i(S_i(t), E(t), A_{-i}(t)) \quad (2)$$

where $S_i(t)$ is the state of agent i at time t , such as the battery level or the consumption and production profiles, B_i is decision rule or policy governing the agent's behavior, as in the case of energy sharing rules provided by regulations or economic and financial incentives, $E(t)$ is the set of the global environment variables, e.g., market electricity prices or costs, and $A_{-i}(t)$ are the action of all other agents except agent i , that influence its decision in terms of possible interactions. Due to this formulation, these models are particularly suited to describing decentralized decision-making in RECs, or to test governance models and incentives.

Simulation results are often fed into optimization models, enabling

model calibration and operational validation and, conversely, optimized results may be used to inform simulations for robustness checks under real-world scenarios [108]. This feedback loop between simulation and optimization enhances model reliability and operational planning and is reported in Fig. 3.

3.5. How can flexibility be incorporated into REC operation?

Answer: Flexibility is essential for balancing variable renewable generation and demand and can be achieved through demand-side management, storage systems, and sector coupling with the main aim of enhancing self-consumption, economic benefits, and grid stability, despite challenges related to user engagement, regulatory barriers and technological constraints.

Flexibility is a general term that refers to the ability of an energy system to adjust energy demand and energy supply in response to specific variations, such as those deriving from electricity prices, grid balances, renewable sources seasonality and intermittency. In the context of RECs, flexibility also consists in maximizing self-consumption and sharing, both from the technical and economic viewpoints. Flexibility can be classified into four main dimensions: temporal flexibility (adjusting loads or generation in time), spatial flexibility (leveraging distributed resources across locations), sectoral flexibility (shifting energy use across electricity, heating, cooling, and mobility), and market-based flexibility (participating in energy markets). Across the dimensions described, different flexibility mechanisms can be put into place. Demand-side flexibility (DSF) is one of the foundational pillars of REC operation, enabling users to modify their consumption behaviors in response to external variations such as electricity prices [109], or grid constraints [110]. In some cases, this can be achieved automated demand response (ADR), i.e. using controllers to regulate building energy systems, such as HVAC, lightning, and domestic appliances in real-time [111]. When smart appliances, such as heat pumps, washing machines or EVs work as flexible loads, i.e. allowing REC members to shift consumption without compromising comfort or functionality, a significant reduction in peak demand, as well as improved grid stability, is usually observed [112]. However, a key challenge for an effective implementation of DSF is properly related to user engagement. As an example, if on one side if real-time tariffs and pricing are crucial to access financial incentives, on the other side the implementation of these strategies often relies on the level of interaction between the consumers and the purpose-built mobile applications or platforms as well as on effective awareness of which kind of benefits may derive from energy savings. In addition, empowering REC members with these digital tools also poses privacy concerns on how these data are effectively handled [113]. This especially applies for vulnerable users, for whom also the access and full use of these technologies can be considered as a barrier.

Storage-based flexibility further complements DSF, having batteries acting as a sort of buffer between generation and consumption, being able to store excess renewable energy generation that can be distributed for later use. Batteries are particularly effective in handling intra-day variability from solar PV systems [114]. Thermal energy storages provide another layer of flexibility by decoupling heating and cooling demands from energy generation and can be especially effective when coupled with solar thermal systems or heat pumps, thus enhancing the overall system efficiency [66]. Another diffused research topic when referring to RECs and flexibility consists in the analysis of vehicle-to-grid (V2G) solutions, where electric vehicles serve as mobile storage units [115]. V2G enables bidirectional energy flow, allowing EVs to discharge electricity to the grid or to the building. This approach not only supports grid services such as peak shaving and frequency regulation but also offers economic benefits to EV owners participating in flexibility markets [116]. It is worth remarking, however, that V2G implementation faces several barriers including the regulatory authority authorization, degradation aspects and also technological hurdles, as they need bi-directional chargers [117].

On the other hand, sector coupling represents a more systemic

Table 2
Comparison of the most diffused optimization methods.

Optimization method	Pros	Cons	Application in the RECs domain
Mathematical programming	High accuracy, globally optimal	Computationally expensive for large systems	Energy dispatch, investment planning
Metaheuristic algorithms	Handle complex, non-linear problems	No guarantee for the global optimum	Microgrid optimization, flexibility management
Stochastic optimization	Account for uncertainty	Requires extensive scenario generation	RES integration, demand response planning
Multi-objective optimization	Handles multiple conflicting objectives	Requires tuning of parameters	Multi-objective REC design
Deep Learning-based optimization	Learns dynamic control policies	Needs extensive training	Real-time REC operation optimization
Hybrid methods	Combines strengths of multiple methods for better performance	Higher computational complexity, requires refinements	Large-scale REC planning, flexibility management

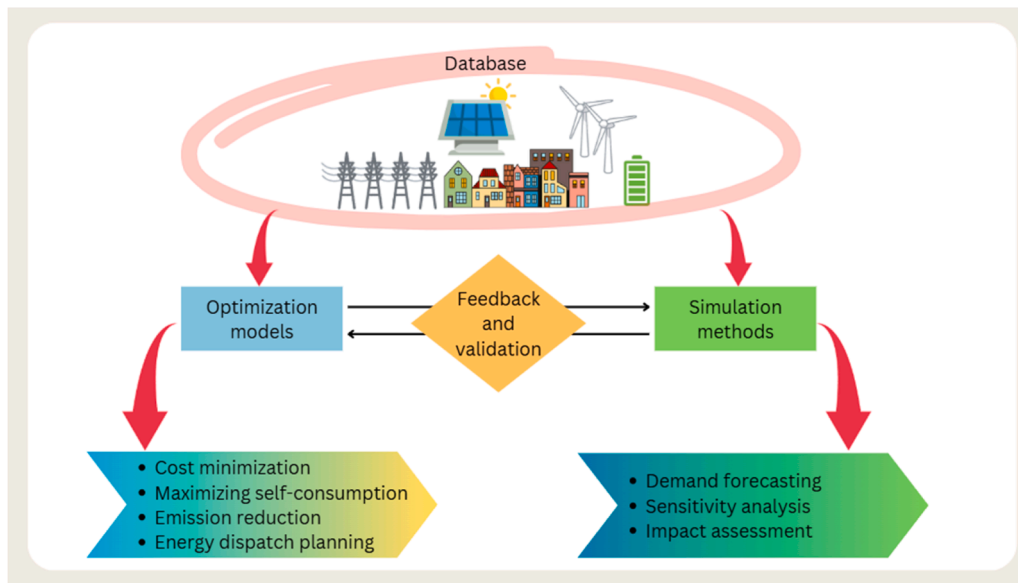


Fig. 3. Workflow of optimization and simulation approaches in RECs.

approach to flexibility by integrating electricity produced by RECs with other energy domains, primarily those belonging to heating and cooling and transportation [118] with the main aim of increasing self-consumption and reducing waste. For instance, P2H technologies, such as electric boilers or heat pumps, can be activated during periods of excess renewable generation, converting electricity into usable heat. Similarly, P2G systems, while less mature and diffuse, especially for RECs, are a promising strategy for long-term storage, such as in the case of hydrogen utilization. Finally, even if not properly fulfilling REC definition, renewable and fuel-based systems are also studied as a hybrid solution to foster the transition toward energy communities, especially in rural areas [119].

Table 3 below summarizes the most diffused strategies for sector coupling, posing the focus on how to handle excess electricity from RECs.

As a result of the discussion so far conducted, Fig. 4 is presented. Specifically, the matrix reports the key flexibility mechanisms for RECs (columns) across the four dimensions (rows). To identify the specific

Table 3
Sector coupling strategies for RECs.

Strategy	Conversion	Characteristics	Use cases
Power-to-Heat (P2H)	Electricity → Heat	Increases self-consumption, enables demand shifting, supports district heating networks	Electric boilers, heat pumps, thermal storage integration
Power-to-Gas (P2G)	Electricity → Hydrogen	Facilitates long-term energy storage, decarbonizes gas use, supports mobility	Hydrogen production, injection into gas grids. Hydrogen vehicles
Vehicle-to-Grid (V2G)	EV battery → Grid	Enhances demand response capacity, provides ancillary services, monetizes EV ownership	EV charging infrastructure, smart charging stations, grid balancing markets
Combined Heat and Power (CHP) + Battery Systems	Fuel/PV → Electricity + Heat	Improves overall efficiency and reliability	Hybrid microgrids, energy communities with district systems
Power-to-Cold	Electricity → Battery	Shifts cooling loads, reduces peak demand in summer	Commercial refrigeration, cooling

contributions of each mechanism, three different levels, i.e. low/medium/high have been used to qualitatively characterize the assignments. A high contribution level indicates that the solutions has a significant impact on the specific solution. As an example, batteries highly impact on the temporal flexibility dimensions, as they are specifically integrated within RECs to disaggregate consumption and production in different intervals. They also have a medium contribution in terms of market impact, as they permit (through the aforementioned disaggregation) to benefit from incentives and tariffs. On the other side, batteries have a low contribution in terms of spatial and sectoral flexibility.

3.6. How can monitoring and energy management be implemented in RECs?

Answer: Robust monitoring and energy management systems for RECs should integrate smart meters, IoT sensors, cloud platforms, and energy dashboards to support peer-to-peer trading, enhance flexibility and meet sustainability goals.

Robust monitoring and energy management tools are essential for real-time performance control of renewable energy assets [120], optimized energy flow management to balance supply and demand [121] predictive maintenance [122] as well as to foster user engagement [123] and demand-side strategies [124]. Effective monitoring systems integrate hardware, software and analytics to establish a consistent operating framework, as reported in Fig. 5.

Smart meters and IoT sensors are deployed across the REC to collect granular data. In the case of smart meters, data can refer to electricity consumption and production at the single building level or at the community level (for example in the case of centralized renewable production plants) and therefore enabling net metering, and also billing. IoT sensors, instead, provide contextual environmental and technical data, such as temperature, irradiance, occupancy, as well as fault detection and equipment monitoring. This data is transmitted in real time to edge controllers, for a pre-processing stage and implementation of preliminary control actions (e.g., managing battery charge/discharge or operating HVAC systems). Data from the edge controllers flows through gateways or routers, which handle communication with higher-level platforms. These devices manage network traffic and apply security protocols to ensure the integrity and privacy of transmitted data. At this level, data is then transmitted to a centralized cloud platform, where it is aggregated, stored, and processed using data analytics and optimization algorithms. Processed information is visualized through a user interface,

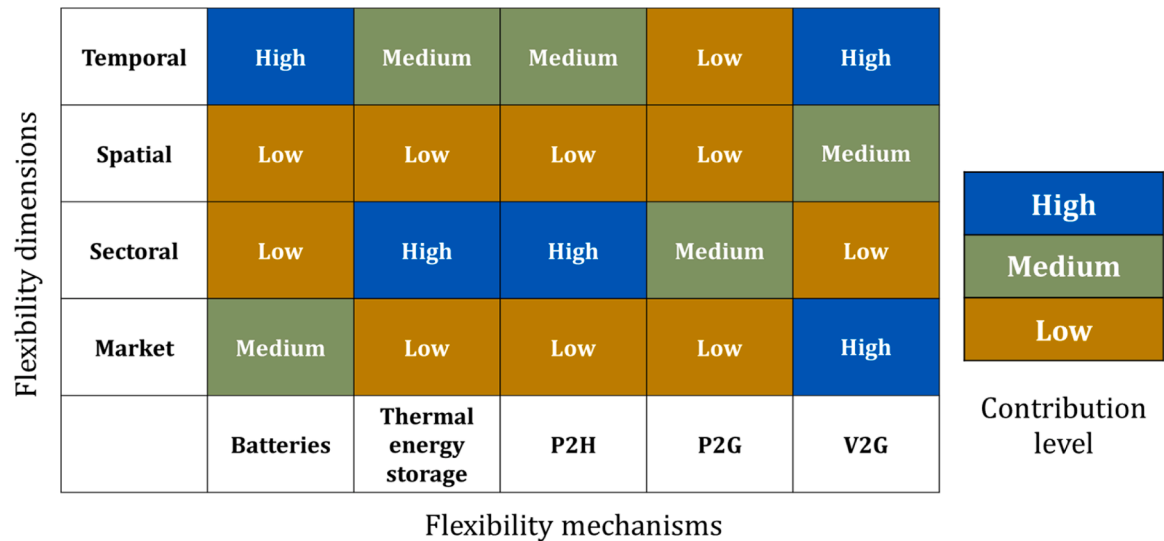


Fig. 4. Flexibility mechanisms in RECs across four dimensions.

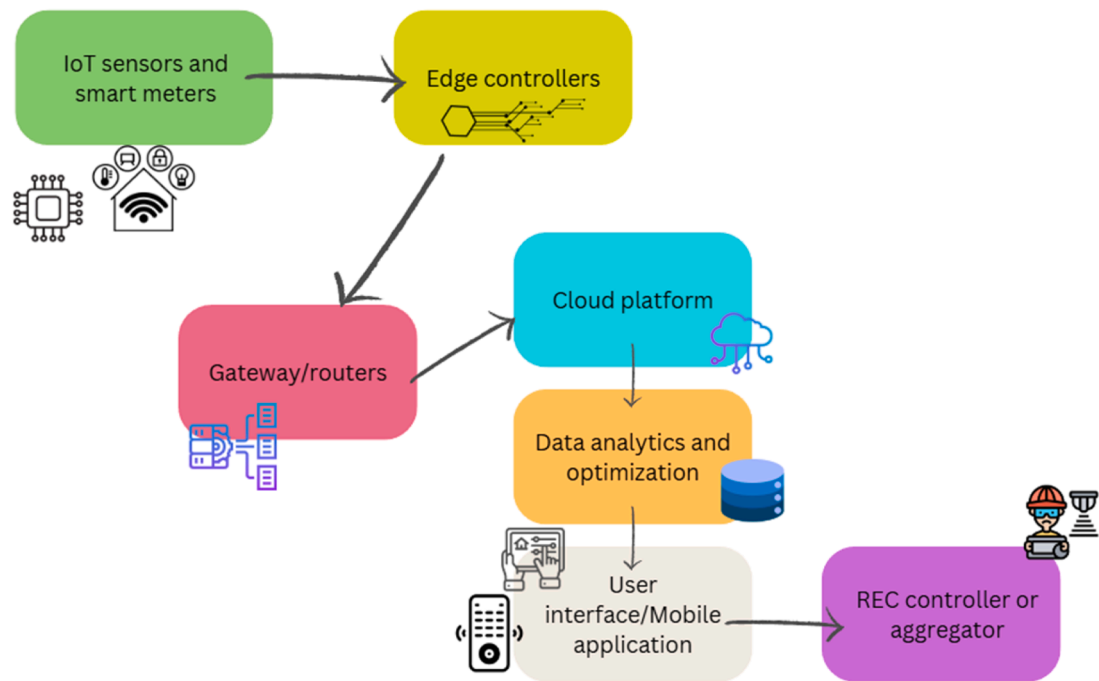


Fig. 5. Architecture of a cloud-based REC monitoring system.

typically an energy dashboard accessible via web browsers or mobile apps and in which energy consumption, generation trends, and financial savings, as well as alerts in case of anomalies can be seen. Interfaces are designed for both individual REC members and administrators, with different levels of access and control, to supervise community-level performance, configure optimization routines, and interact with grid operators or markets.

Efficient energy management ensures RECs meet economic, environmental, and social goals, one of the key features of REC included in the definition provided by the RED II Directive [9]. In this context, different strategies can be implemented such as the above-discussed general flexibility mechanisms. In case of DR mechanisms, when dealing with energy management, three different programs can be identified, as reported in Table 4, along with the specific mechanism they refer to and the benefits and barrier associated to each typology.

Table 4 DR mechanisms.			
DR Typology	Mechanism	Benefits	Challenges
Price-based DR	Dynamic tariffs encouraging off-peak consumption	Peak load reduction	Smart meters integration is a must
Incentive-based DR	More incentives are received for the compensation	Grid stability	Complex management and participation
Automated DR	AI-controlled energy devices are installed to respond to grid signals	High efficiency	Cybersecurity risks, high costs for implementing the communication infrastructure

The three DR typologies differ in how they influence energy

management. Price-based DR shifts consumption through dynamic tariffs, encouraging REC members to adjust their energetic behavior [125]. Incentive-based DR implies direct financial revenues and is usually coordinated by DSO Automated DR, instead, adopt AI-techniques without any direct involvement of members [126]: in this case, however, and more than the others, such systems demand secure, interoperable communication networks and raise concerns regarding data protection and user privacy [127].

Regarding the management of the energy flows within a REC, peer-to-peer (P2P) energy trading enables members to exchange surplus energy locally, promoting decentralization and energy autonomy [128]. Several mechanisms support P2P trading, each with distinct characteristics. These mechanisms could involve in the near future market-based models [129] bidding strategies [130] and decentralized automation platforms with robust IT infrastructure [131].

Each of these approaches contributes to increasing the management and scalability of REC energy markets [132], although they require different levels of technical maturity and regulatory supports [133].

3.7. To what extent can RECs contribute to the achievement of sustainable development goals?

Answer: RECs play a pivotal role in advancing multiple SDGs, by promoting local renewable energy generation, reducing energy poverty and carbon emissions, fostering sustainable urban development, contributing to economic growth and enhancing social equity and access to clean energy.

The United Nations Sustainable Development Goals (SDGs), adopted under the 2030 Agenda for Sustainable Development, provide a global framework of targets and indicators to address interconnected economic, social, and environmental challenges [134]. Among these, the energy transition plays a cross-cutting role. RECs, as decentralized urban energy forms in which prosumers and consumers engage in renewable energy production, consumption, and sharing, represent a key innovation to target this transition. Their deployment inherently aligns with several SDGs, particularly those concerning energy poverty (SDG 1 “End poverty in all its forms everywhere”), clean energy access (SDG 7 “Ensure access to affordable, reliable, sustainable and modern energy for all”), economic growth and decent jobs (SDG 8 “Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all”), innovation (SDG 9 “Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation”), reduction of inequality (SDG 10 “Reduce inequality within and among countries”) sustainable urban development (SDG 11 “Make cities and human settlements inclusive, safe, resilient and sustainable”), sustainable consumption and production (SDG 12 “Ensure sustainable consumption and production patterns”), and climate action (SDG 13 “Take urgent action to combat climate change and its impacts”).

A thorough evaluation of RECs within the SDG framework requires translating community-level performance metrics, such as self-consumption rates, grid balances, emission rate, into SDG-relevant indicators. Some literature focused on similar topics [135]. Still, there are open questions on how these indicators can be tailored to specific urban forms.

RECs are strictly connected with SDG7, as they address energy affordability and sustainability through local generation from renewable sources, typically solar PV, small wind turbines, and biomass. By reducing reliance on centralized grid-supplied electricity, RECs lower retail electricity costs, especially when combined with self-consumption optimization strategies. Different studies indicate that renewable-based systems can reduced household electricity costs in comparison with conventional supply [136,137].

By facilitating high penetration of RES at the community level, RECs significantly reduce carbon emissions from electricity use, as part of the targets included in SDG 13. Emissions savings in RECs is also significant, especially if coordinated with operational control techniques [138]

carbon pricing [139] environmental patents and energy efficiency [140] ecological footprint of energy consumption [141].

Being sustainability part of their definition, RECs directly support the development of sustainable urban energy infrastructures for cities and communities, as envisaged in SDG 11. By enabling localized production and flexible consumption, RECs reduce the load on central distribution [142].

At the same time, RECs also have an impact with respect to SDG 8 and SDG 9, regarding the economic growth and innovation, as their diffusion stimulates local economic activity through construction, installation and operation of distributed energy assets [143]. According to IRENA, energy communities may contribute to the creation of direct and indirect jobs, including roles in technical services, data analytics, and customer support [144]. RECs may also act as incubator for innovation, especially in the case of integrating digital technologies for energy trading, AI-based control, digital twin for real-time operation.

RECs also contribute to the achievement of SDG 1 and SDG 10, by fostering social equity due to the democratization of the access to clean energy and enabling vulnerable populations to participate in the energy transition. Mechanisms such as incentives and cooperative ownership models ensure inclusion of low-income households, renters and marginalized groups [145]. Beyond income considerations, RECs are also relevant to energy poverty, defined as the lack of affordable, reliable, and modern energy services, which affects rural households, elderly individuals, and low-income dwellers [54].

3.8. Can RECs be scaled to the city and neighbourhood level?

Answer: RECs can vary in size depending on regulatory frameworks and urban features. From small configurations it is possible to evolve to larger projects, such as PEDs that integrate electricity, thermal and mobility demands, although more complex configurations require careful planning and can reduce flexibility for member.

RECs do not have to respect a standardized size, rather, the specific applications and regulatory frameworks within each country often influence their dimensions. These frameworks may impose certain guidelines that contribute to the variance in extension. For example, in Spain and France, regulations establish a specific maximum distance between REC members and the generation plant. Conversely, in other States, energy sharing is typically confined to the boundaries of the same electrical substation [52]. In this context, a thoughtful analysis of the urban area holds significant importance. It is worth noting that the REC may vary in size relatively to a municipality, reflecting into the diverse characteristics of small villages as well as larger cities. In the first case, it could be possible for the entire population to participate, which is likely to respect to the established constraints. In the second case, it may verify that a REC has a maximum allowable size of a city neighbourhood. In this instance, specific features can play a crucial role in shaping the project, as investigated in the study [146] that compares three districts within the same city, each exhibiting unique characteristics related to built-up areas and open spaces. To achieve an equivalent size of PV system, the option of utilizing one block of buildings in the district with higher availability of public buildings has been considered, while considering seventeen small houses situated in that one characterized by lower housing density and fewer public buildings. In order to enhance the benefits for REC members, this analysis has been concentrated on a specific segment of each district for the REC implementation, conveniently accessible by foot or bicycle within approximately 15 to 20 min from the district's central location [146]. The exploration of low-density communities is focused on attaining 100 % RES targets, recognizing the considerable space available for renewable energy development and the inherent limitations of energy density in these areas [147].

The limited size of a REC can facilitate effective management by fostering collaboration among its members, which in turn can enhance a collective identity and belonging [148], and simplify the legal management of the configuration. For a REC that consists of a condominium

with a rooftop PV plant and BESS, applying an optimal control strategy becomes more straightforward [24]. The relatively compact dimensions facilitate a more thorough understanding of the technical characteristics and energy consumption of each building involved [20]. However, scaling down the configuration emphasizes the importance of involving individuals with the appropriate expertise in each project to ensure effective energy and economic management. In certain contexts, such as Italy, the regulatory landscape evolved over time, enabling the establishment of RECs at both market-area and national levels. Furthermore, this framework allows for the assessment of shared energy by considering the specific boundaries of each HV/MV electrical substation [149]. The intermediate scale of urban neighbourhoods presents a valuable opportunity to harmonize energy supply and demand while encouraging the development of high-performance buildings and districts [150]. The involvement of people who live in the same district and who constitute a CER, allows also to promote the achievement of social goals very felt in that area, such as the fight against energy poverty in social housing district [32]. In the future, RECs can evolve in city districts or suburban areas. Smart cities resemble smart communities but operate on a larger scale, featuring unique aspects, especially concerning involved stakeholders and their interaction [151].

In this framework, the REC can move towards the concept of positive energy districts (PED). The latter is defined as a mixed-use area that emphasizes high energy efficiency, with the goal of achieving net zero CO₂ emissions and generating a positive energy surplus from RES on an annual basis [152].

Although these entities may not have formal legal recognition, there is a growing interest in them within the research community, since they include interaction among users, building and mobility ensuring sustainability benefits, which thus allows for a broader range of potential applications compared to RECs [153]. The achievement of the PED targets represents an evolution of what a REC already proposes to do, considering an expansion of energy demands that in addition to electricity also considers thermal and cooling energy demand [154].

The inclusion of thermal energy sharing is a key factor that can drive future community projects. In this scenario, members of the same REC need to have a supportive infrastructure, such as a DHN, which enables them to connect physically and utilize thermal energy effectively [79]. This is certainly a significant evolution of configurations but on the other hand it leads to greater rigidity as users have less flexibility to alter their settings. Similarly, new users would be barred from joining unless a pre-planned connection is made. Regarding size, the REC here would be much more site-specific, and a reduction or enlargement would not be easy to implement.

3.9. What business models can be applied to RECs?

Answer: Business models for RECs should balance economic viability, social inclusiveness, and environmental goals, with strategies varying based on local legal, technical and economic conditions. From the simple self-consumption scheme to multi-sector integration and related investment models the focus should be on the creation of shared value beyond financial returns.

Business models are strategies that can be implemented for enhancing the profitability of any company or organization. In the case of RECs, it is important to implement the recommendations included in the RED II definition, i.e. RECs should not be conceived as urban forms merely dedicated to self-consumption and energy sharing. On the contrary, they should promote environmental, economic and social benefits to their members and, in general, to the urban environment in which they operate. The focus for community-based projects is on creating value through aggregating stakeholders, as well as managing and allocating resources and activities. The REC model, therefore, is not just a vehicle for distributing clean energy, but a broader framework for distributing benefits and responsibilities among community stakeholders [12]. This is a new way of thinking, especially considering the

traditional and centralized energy sector, in which the maximization of the profits was at the center of any decision-making process. Now, REC models must integrate economic viability with social inclusiveness, democratic governance, and environmental performance [155].

However, and also discussed in question before on policies and regulations, one of the fundamental challenges in designing REC is the heterogeneity of legal, technical, and economic conditions across Europe as well as the different granularity level [156], and this impact also the definition of business models. Key factors influencing business models include members typology, sharing model, revenues streams, key activities, resources and partners, that can significantly vary case-by-case [157]. For this reason, the operation context of each REC can lead to prefer a model rather than another one. Many RECs are built around a single energy vector, typically solar PV for self-consumption [68]. In such cases, members aim to lower their electricity bills by reducing their dependency on the grid [158]. However, participation in RECs offers fewer economic benefits than existing social tariffs or general subsidies, especially for low-income households and, therefore, the participation of members may be not attractive [159].

A promising evolution involves coupling electricity production and sharing with services such as sector coupling [160], energy efficiency retrofits and social impacts of operation [161]. These approaches enable RECs to better align energy supply and demand within their boundaries and generate additional revenue streams. A further extension of the business model involves participation in local flexibility markets, with RECs acting as aggregators of demand-side flexibility or distributed storages [162].

The way RECs are constituted has also an important impact on the business models. RECs may be initiated by local authorities, citizen cooperatives, energy service companies, or hybrid consortia, thus influencing both cost structures and income potential. In areas with strong municipal involvement, such as in the case of the Italian regulations due to specific incentives dedicated to small cities, RECs often benefit from existing administrative support and infrastructure, facilitating scale and continuity [163]. In contrast, communities relying solely on personal efforts may face limitations in professionalization, maintenance, and growth.

To navigate the diversity of business logic, five typologies of REC business models can be identified. These models, compared in Table 5, differ in terms of revenues, technical requirements, scalability, and vulnerability to policy or market changes. Across these categories, the evolution from a pilot to a fully operational REC entails not only technical readiness but also financial planning, regulatory compliance, and community engagement.

Economic considerations are often a priority in various contexts, but they typically support only one of the targets of RECs, which is primarily driven by sustainability. Indeed, the dimensions of sustainability focus more on the environmental and social aspects rather than the economic one. Therefore, the members of the RECs and their effective management aim to create co-value that aligns with the community's complex needs, ensuring that the successful business models must reconcile economic, technical, and social dimensions [164]. They must be adaptable to different local contexts, resilient to changes in energy regulation, and transparent to foster trust among members. The diversity of possible models is not a limitation but a strength—provided there are adequate policy signals, institutional support, and financial mechanisms to ensure that communities can choose and refine the path that best fits their collective vision and capacity [165].

3.10. Which policies have been implemented to support REC development?

Answer: RECs development is rooted in the RED II Directive and supported by various EU and national frameworks. Implementation, however, is still uneven, highlighting the need for greater harmonization, clear rules and improved monitoring.

Table 5
Comparison of emerging REC business model types.

Model type	Revenue sources	Key stakeholders	Complexity	Scalability	Barriers
PV self-consumption	Electricity savings	Households, cooperatives	Low	Medium	Tariff regulation, limited financial gain
Multi-sector integration	Energy sales (heat, mobility), efficiency	Municipalities, mobility providers	Medium	High	Technological and organizational coordination
Flexibility aggregation	Grid services, demand response	Aggregators, DSOs, technology providers	High	Medium/High	Regulatory complexity, IT infrastructure high costs
Community investment	Community capital returns (dividends)	Cooperatives, financial entities	Medium	Variable	Capital intensity, financial literacy
Social-oriented model	Subsidies, social tariffs, grants	Municipalities, NGOs, citizen no-profit organizations	Low	Context-based	Policy dependency, revenue instability

The establishment of RECs in Europe is rooted in the European Union's Clean Energy for All Europeans Package [166] particularly in the Renewable Energy Directive (EU) 2018/2001, commonly referred to as RED II [9]. The Directive formally defines the roles and rights of RECs, legally recognizing them as non-commercial legal entities with autonomy, voluntary participation, and democratic governance. In addition, the Directive also specifies that RECs should have as primary objectives to bring environmental, economic and social benefits to members, an element that also has to be included into each national transposition. In addition, the Directive obliges Member States to create enabling frameworks that remove unjustified regulatory and administrative barriers, facilitate access to financing, and allow RECs to engage in generation, consumption, storage, and energy sharing. Despite these indications, however, the transposition varies widely across Member States [10].

Along with the national regulations, the European Commission supports RECs development also through guidance documents and funding programmes. Notably, the Energy Communities Repository [167] and Rural Energy Community Advisory Hub [168] are platforms launched by the EU to provide technical assistance, facilitate knowledge exchange, and support pilot initiatives. These tools are complemented by structural and investment funds, such as those under the Just Transition Mechanism [169] and Horizon Europe [170]. Moreover, Horizon 2020 and Horizon Europe projects have funded demonstration actions, capacity building, and digital innovation in RECs [171].

Financial support for RECs in Europe is anchored in both EU-level and national instruments. At the EU level, the European Regional Development Fund (ERDF) [172] and the Cohesion Fund [173] have been instrumental in co-financing REC infrastructure, particularly in underserved regions.

At the national level, Member States have adopted diverse approaches to financial incentivization. Austria is a front-runner in terms of number of established energy communities [174]. Here, electricity sharing is regulated by the Electricity Act, e Elektrizitätswirtschafts- und organisationsgesetz EIWOG [175] and by the Ökostromgesetz ÖSG, referring to RECs [176]. Austrian law recognizes different REC models: local, regional and nationwide communities, depending on the voltage network used for sharing. The regulation recognizes 57 % discount for using the same low voltage network for local REC and a discount of 28 % in the case of regional REC in low voltage and 64 % for regional REC in medium voltage.

REC in Italy were first introduced through the Decreto Milleproroghe 2020, Law n. 162/2019 [177] which can be considered as the pilot phase for REC implementation. Subsequently, the Italian Regulatory Authority for Energy, Networks and Environment, ARERA, issued the Resolution 318/20 to regulate the tariff and incentives [178] while the GSE (Gestore Servizi Energetici) published the technical rules [179]. Later, ARERA's 72/2022/R/EEL regulated the self-consumption schemes [180] and now the Italian Ministerial Decree 414 of December 2023 (also known as Decreto CER) regulates the final incentive and tariff premiums for RECs [181]. According to the Decreto, feed-in tariffs are set at up to 110 eur/MWh for shared renewable energy within Italian REC. In addition, a capital contribution of up to 40 % of

the investment cost of the plant may be covered by available funds from the NextGeneration EU program. In France, RECs are governed under the framework of collective self-consumption, primarily regulated by the Code de l'énergie [182]. The document Arrêté du 6 octobre 2021 [183] defines technical and administrative procedures for energy sharing, while the Arrêté du 21 novembre 2019 [184] sets the financial conditions and tariff structures for collective self-consumption operations. Germany's Erneuerbare-Energien-Gesetz (EEG) has historically supported energy cooperatives through guaranteed remuneration mechanisms [185]. In Spain, the Royal Decree 244/2019 introduced a regulatory framework enabling collective self-consumption with net metering benefits [186]. The Czech Republic is piloting REC schemes under the Energy Act amendment, also known as Lex OZE III, with a focus on multi-apartment building collectives [187]. Romania has adopted Law 123/2012 (amended 2022) to facilitate energy sharing and prioritizes vulnerable consumers [188]. A summary of the various regulatory declinations in different EU countries and related incentives is proposed in Table 6.

RECs also intersect with the EU's goals on social inclusion and energy justice. Article 22 of RED II emphasizes the participatory and equitable nature of RECs, encouraging the inclusion of vulnerable households and municipalities [9]. Despite the indication, equity-based instruments remain inconsistently implemented across Member States. There is a growing consensus among experts that inclusion criteria, governance transparency, and benefit-sharing rules must be codified in national frameworks to prevent elite capture and exclusion [189].

While many policy mechanisms have been launched, a comprehensive assessment of their effectiveness remains lacking. There is a need for EU-wide databases to track REC performance, regulatory maturity, and socio-economic impact, as this would be a powerful tool for any stakeholder involved in the REC design and implementation process [174]. In parallel, there is room for enhanced legal clarity on the interaction between RECs and DSOs, particularly regarding balancing responsibilities and cost allocation.

4. Future directions

The development of RECs has the potential to significantly reshape the energy landscape. However, to fully harness this capability, there is the need to address a series of challenges that can be directly linked to opportunities for the future, in terms of research, policy and practice.

From the urban and energy planning perspectives, RECs are expected to evolve into more complex configurations, such as PEDs or contribute to the concept of climate-neutral cities advocated by the European Union. This will require a net advancement in terms on energy management and control, especially when dealing with multi-vector energy systems and smart district infrastructures.

Future RECs will likely rely on dynamic and real-time data with dedicated analytics and operational models. To this scope, a significant attention, finally, should also be devoted to the digitalization and exploitation of Artificial Intelligence (AI) technologies. In this context, digital twins, edge computing and blockchain will increasingly support production, consumption and distribution of energy, as well as

Table 6

Summary of key policies and regulations in EU Member States.

Country	Regulation	REC types/scopes	Financial support/incentives
EU-wide	RED II (Directive 2018/2001)	Legal definition of RECs	ERDF; Horizon Europe; Cohesion Fund; Just Transition Mechanism
Austria	EIWOG, ÖSG	Local, regional, national RECs based on grid level	Up to 64 % grid fee discount (regional MV), 57 % (local LV), 28 % (regional LV)
Italy	Law 162/2019, ARERA Resolutions 318/2020 and 72/2022, Decreto CER	Local RECs (within HV/MV substations)	Feed-in tariff up to 120 €/MWh, plus 40 % investment cost coverage
France	Code de l'énergie, Arrêté 6 Oct 2021, Arrêté 21 Nov 2019	Collective self-consumption RECs	Tariff structures and administrative procedures defined by national decree
Germany	Erneuerbare-Energien-Gesetz (EEG)	Energy cooperatives	Guaranteed remuneration mechanisms
Spain	Royal Decree 244/2019	Collective self-consumption	Net metering benefits
Czech Republic	Amendment to Energy Act (Lex OZE III)	Multi-apartment buildings collective	Pilot-phase incentives under development
Romania	Law 123/2022	Energy sharing	Measures for vulnerable households

predictive maintenance or peer-to-peer optimal transactions.

In terms of social innovation and governance models, future RECs will emphasize the role of capacity building and participatory decision-making, especially engaging vulnerable populations and promote energy justice. This can be achieved only with a solid policy harmonization among different legislations, with the possibility for any stakeholder to have access to databases, tools and models to support RECs diffusion from the preparatory design to the effective constitution, operation and monitoring. To achieve this goal, however, common standards for data exchange and systematic integration should also be defined to help ensuring compatibility across countries, platforms or technologies.

Thus, future directions of RECs will depend on the extent to which technological innovation, social inclusion and regulatory aspects will be efficiently integrated into a systemic framework, capable of responding to different challenges of energy transition and generating shared value at both local and European scales.

5. Conclusions

Renewable Energy Communities are important paradigms within the energy transition, as they enable citizens and local actors to actively manage their energy consumption, production and sharing. Their successful implementation, however, hinges on the different aspects such as technical, regulatory, and organizational strategies. To provide a structured understanding of RECs, ten questions have been formulated, starting from the foundational aspects.

Regarding the membership that typically characterized RECs, there is evidence that various stakeholders, ranging from individuals to public bodies, can foster their development. In this sense, the configuration of a REC strongly depends on the members involved in its constitution, especially in terms of democratic participation and social cohesion.

The status of RECs' development is, however, very heterogeneous, with regulatory maturity and incentive schemes that vary significantly across countries and regions, from European nations characterized by mature implementation frameworks to the international context, where RECs can be found as embedded in microgrid or local sustainability models. The analysis on the REC status is definitely crucial for tailoring practices and identifying replicable solutions.

In terms of key technologies that enable the REC diffusion, it can be stated that PV remain dominant, due to its scalability and modularity. Other renewable (wind, biomass, thermal solar) and hybrid energy systems (cogeneration) are increasingly taking the scene, and this is more appreciated for multi-sector integration. Optimization and simulation tools permit the planning and operation of RECs, enabling multi-objective decision-making, performance assessment and dynamic interaction with the energy markets. In parallel, robust monitoring and energy management infrastructures are essential to ensure the real-time control and optimization of energy flows. In the context of RECs, flexibility strategies are going to be considered crucial as the allow matching variable supply with dynamic demand. This flexibility spans demand

response mechanisms, storage integration, sector coupling, and peer-to-peer trading, each contributing to increased autonomy and reduced grid dependency, as well as increased benefits for the members. In addition, as flexibility services become more digitized and responsive, the role of AI, digital twins, and predictive analytics will grow, expanding the ability of RECs to participate in ancillary markets and provide grid-supportive services.

RECs can significantly contribute to the SDGs, especially in targeting the objectives related to energy poverty reduction, inclusive infrastructure, clean energy production and climate impacts mitigation. This also paves the way for scaling RECs from individual buildings to neighborhoods or city districts. While technical and regulatory constraints exist, such as those arising on grid proximity and interconnection rules, RECs effectively have a great potential to evolve into PEDs, able to combine thermal, electric and mobility networks into integrated urban energy ecosystems. Equally important is the development of tailored business models. RECs require structures that are not only financially viable but also socially inclusive and operationally replicable. Emerging models range from self-consumption schemes to service-based aggregators, each shaped by local governance, investment capacity, and regulatory frameworks. And, specifically pointing to regulations, the transposition of the RED II in each Member State remains uneven, and harmonized enabling conditions are still needed to unlock the full potential of RECs.

CRediT authorship contribution statement

A. Fichera: Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **E. Marrasso:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **C. Martone:** Writing – review & editing, Writing – original draft, Visualization, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **G. Pallotta:** Writing – review & editing, Writing – original draft, Visualization, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **M. Sasso:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **R. Volpe:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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